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Your Name*
Your Affiliation

Co-author Name
Co-author Affiliation

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Abstract

1 Introduction

The rapid advancement of artificial intelligence has led to the deployment of highly capable models across numerous domains. However, these systems often make decisions for reasons that remain opaque, introducing novel risks regarding trust, safety, and alignment Cunningham et al. [2024]. To mitigate these risks, the field of Mechanistic Interpretability aims to reverse-engineer neural networks by breaking them down into smaller, analyzable computational units Elhage et al. [2021].

In the domain of computer vision, Convolutional Neural Networks (CNNs) have historically been the de-facto standard. Recently, Vision Transformers (ViTs) Dosovitskiy et al. [2021] have achieved comparable or superior performance, highlighting a fundamental question: how do Vision Transformers solve visual tasks? As noted by Raghu et al. Raghu et al. [2021], ViTs do not simply learn convolutional inductive biases from scratch; instead, they rely heavily on self-attention mechanisms to enable the early aggregation of global information across spatial locations, strongly propagating features through residual connections.

Understanding these internal representations requires analyzing individual neurons. However, a major roadblock in mechanistic interpretability is *polysemanticity*—the phenomenon where individual neurons activate in multiple, semantically distinct contexts. Recent frameworks Elhage et al. [2021] hypothesize that this arises from *superposition*, where networks represent more features than they have dimensions by packing them into an overcomplete set of directions in the activation space. Recently, Sparse Autoencoders (SAEs) have emerged as a powerful tool to untangle this superposition, learning sets of sparsely activating features that are highly interpretable and monosemantic Cunningham et al. [2024], Bricken et al. [2023].

Despite these theoretical breakthroughs, the application of mechanistic interpretability remains heavily biased towards Large Language Models (LLMs). This disparity leaves significant research gaps in the visual domain:

- **The Spatial Challenge:** Unlike language models that process sequential word tokens, the spatial structure of ViTs (processing 2D patch tokens) and their bidirectional attention mechanisms create unique, under-explored challenges in tracking the causal flow of information.
- **Cross-Modal Evaluation:** While SAEs have successfully decomposed MLP activations into interpretable concepts in LLMs, applying this to ViTs requires mapping visual latent spaces to human-understandable concepts, necessitating novel cross-modal evaluation strategies.

*Corresponding author: `your.email@example.com`

In this project, we address these gaps by applying mechanistic interpretability techniques to a pre-trained Vision Transformer. By leveraging Sparse Autoencoders, we aim to decompose its internal representations to understand the computational structures that drive its visual predictions.

2 Related Work

This section reviews the foundational concepts and recent advancements that intersect our research: the mechanistic interpretation of transformer architectures, the resolution of polysemanticity through dictionary learning, and the analysis of internal representations in Vision Transformers.

2.1 Mechanistic Interpretability in Transformers

The primary goal of mechanistic interpretability is to reverse-engineer the computational components of neural networks into human-understandable algorithms. Elhage et al. [2021] established a mathematical framework for understanding Transformer circuits, conceptualizing the architecture around the *residual stream*. The residual stream acts as a central communication channel, carrying the accumulated representations of tokens across the network. Attention heads and Multi-Layer Perceptron (MLP) blocks independently read from and write to this stream. Specifically, attention heads route information between tokens (spatial locations, in the context of images), while MLP blocks process information within individual token representations. Understanding this residual communication is crucial for determining the exact layers and components responsible for specific model behaviors.

2.2 Sparse Dictionary Learning and Monosemanticity

A significant obstacle in mechanistic interpretability is *polysemanticity*, where individual neurons activate in response to multiple, unrelated features. This phenomenon is largely attributed to *superposition*, a strategy where models represent more features than they have dimensions by embedding them in an overcomplete set of directions within the activation space Elhage et al. [2021].

To untangle these representations, recent works have successfully applied Sparse Autoencoders (SAEs) Cunningham et al. [2024], Bricken et al. [2023]. SAEs aim to decompose the dense, polysemantic activations into a larger dictionary of sparse, *monosemantic* features. The training objective of an SAE typically combines a reconstruction loss (Mean Squared Error) to ensure fidelity to the original activations, and an L_1 regularization penalty to enforce sparsity. By applying SAEs to the MLP layer outputs or the residual stream, researchers have been able to isolate interpretable concepts that directly influence model predictions.

2.3 Internal Representations of Vision Transformers

While Transformers were originally designed for natural language, Dosovitskiy et al. [2021] demonstrated that an image can be processed as a sequence of 16x16 spatial patches, leading to the Vision Transformer (ViT) architecture. However, the way ViTs process visual information fundamentally differs from traditional Convolutional Neural Networks (CNNs).

Raghu et al. [2021] provided a systematic comparison of the internal representations of ViTs and CNNs. They discovered that ViTs exhibit highly uniform representations across layers and heavily rely on self-attention to enable the early aggregation of global information, bypassing the localized receptive fields typical of early CNN layers. Furthermore, the residual connections in ViTs strongly propagate features from lower to higher layers. While these insights clarify the macroscopic behavior of ViTs, the microscopic, circuit-level mechanisms—and how specific visual concepts map to the aggregated [CLS] token—remain largely unexplored compared to their linguistic counterparts.

3 Experiments

4 Results

5 Discussion

6 Conclusion

7 Methodology

Acknowledgments

References

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A Appendix Title